

| Req ID         | Type          | Description                                                                                                                                                                                        | Priority | Acceptance Criteria                                                                                                                                                                                                           | Rationale                                                                                                              |
|----------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>FR-001</b>  | Functional    | The system shall enable physicians to author and digitally sign structured prescriptions that are automatically compiled into downloadable PDF records.                                            | High     | Pass: Valid cryptographic signature attached to prescription PDF.<br><br>Fail: Prescription generated without doctor license validation.                                                                                      | Ensures prescriptions are authentic, tamper-evident, and legally usable at pharmacies.                                 |
| <b>FR-002</b>  | Functional    | The system shall allow patients to view a list of available physicians filtered by specialty, along with each physician's open (unbooked) consultation slots.                                      | High     | Pass: Selecting a specialty filters the physician list correctly and only unbooked slots are shown as selectable.<br><br>Fail: A slot already reserved by another patient appears as available.                               | Patients need accurate, up-to-date availability to make an informed booking decision.                                  |
| <b>FR-003</b>  | Functional    | The system shall allow a patient to book a specific video consultation slot with a chosen physician, producing a confirmed booking record after successful payment.                                | High     | Pass: After booking, the slot status changes to "Reserved," is removed from other patients' views, and appears in the patient's dashboard.<br><br>Fail: The same slot can be booked by two different patients simultaneously. | Prevents double-booking and is the core transaction the platform exists to support.                                    |
| <b>FR-004</b>  | Functional    | The system shall automatically generate and deliver an encrypted, single-use tele-consultation room link to both the patient and the physician no later than 10 minutes before the scheduled slot. | High     | Pass: Link is delivered via app notification/email at least 10 minutes before slot start and becomes invalid once the session ends.<br><br>Fail: The link remains usable after the consultation window has closed.            | Restricts session access to the intended parties and prevents replay/ unauthorized entry into a consultation.          |
| <b>FR-005</b>  | Functional    | The system shall allow a patient to download their digitally signed prescription as a PDF from their consultation history at any time after it has been issued.                                    | Medium   | Pass: The downloaded PDF matches the signed record on file and displays the physician's digital signature and prescription details.<br><br>Fail: A prescription PDF can be downloaded before the physician has signed it.     | Gives patients continued access to their medical records for pharmacy use and personal reference.                      |
| <b>NFR-001</b> | Nonfunctional | All video session metadata and patient clinical notes must be encrypted in transit using TLS 1.3 and at rest with AES-256.                                                                         | High     | Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.<br><br>Fail: Any clinical data is found transmitted or stored without the specified encryption.                             | Protects sensitive patient health data (PHI) from interception or unauthorized access, a legal/compliance requirement. |
| <b>NFR-002</b> | Nonfunctional | The consultation booking and video-session service shall maintain at least 99.5% uptime per calendar month, excluding published maintenance windows.                                               | High     | Pass: Uptime monitoring logs confirm $\geq 99.5\%$ availability over a rolling 30-day period.<br><br>Fail: Unplanned downtime exceeds the 0.5% monthly threshold.                                                             | Patients booking time-sensitive or urgent consultations depend on the platform being reliably reachable.               |